

SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No : 192524108 Name: S. LAKSHITA

COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

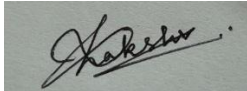
Activity Tool : Visualization Critique & Redesign

Weightage: 20%

Code of Conduct:

I S. LAKSHITA(192524108) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Evaluation of Existing Visualization 25%	Redesign Strategy 25%	Application of Visualization Principles 20%	Organization 15%	Accuracy 10%	Clarity 5%	Total
Q1							
Q2							
Q3							
Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

1) Question 1 – Critique and Redesign an ECDF and Q-Q Plot

The dataset contains **20 employees** from IT, HR, Finance and Marketing.

Assessment Task 1: Create the two original-style plots in R.

```
library(ggplot2)
```

```
salary_data <- data.frame(  
  Department = c(rep("IT",5), rep("HR",5),  
                 rep("Finance",5), rep("Marketing",5)),  
  Salary = c(  
    42000,45000,48000,51000,55000,  
    38000,40000,42000,46000,50000,  
    47000,49000,52000,58000,65000,  
    36000,39000,43000,47000,72000  
  )  
)
```

```
# Original ECDF
```

```
ggplot(salary_data, aes(Salary)) +  
  stat_ecdf(aes(color=Department)) +  
  labs(title="Salary Distribution",
```

```

x="Salary", y="Value") +

theme_dark()

# Original Q-Q Plot

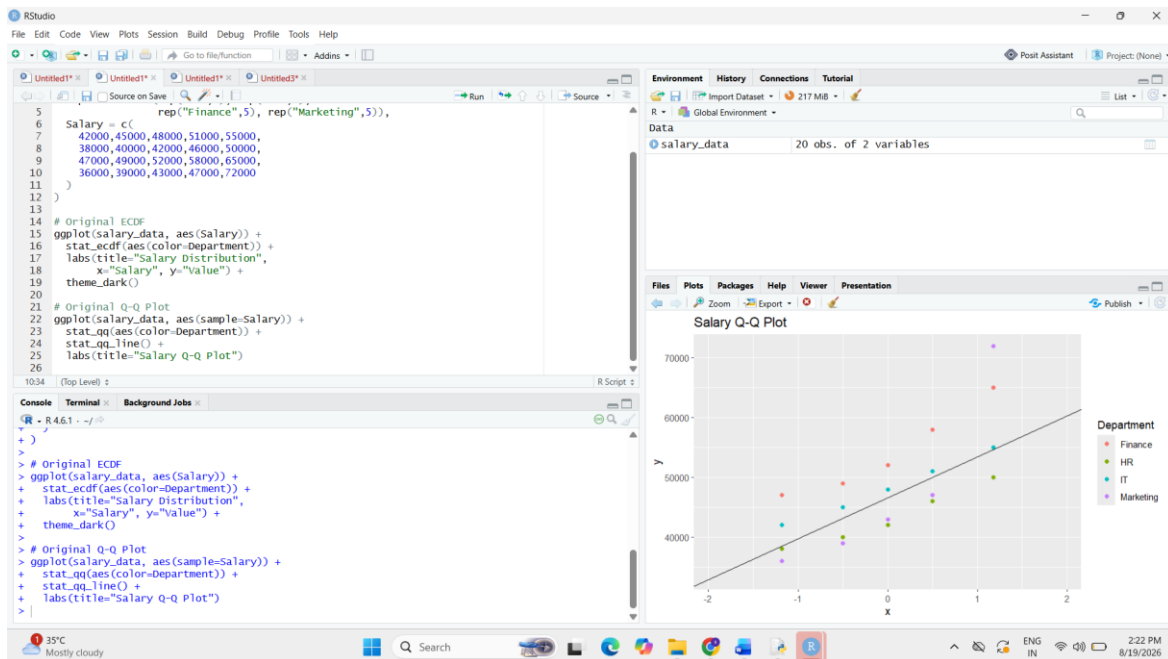
ggplot(salary_data, aes(sample=Salary)) +

stat_qq(aes(color=Department)) +

stat_qq_line() +

labs(title="Salary Q-Q Plot")

```



Assessment Task 2: Visualization problems

1. Unclear y-axis label: "Value".
2. Excessive decoration and dark theme.
3. Unnecessary colour variation.
4. Poorly explained salary scale.
5. Labels are not sufficiently informative.

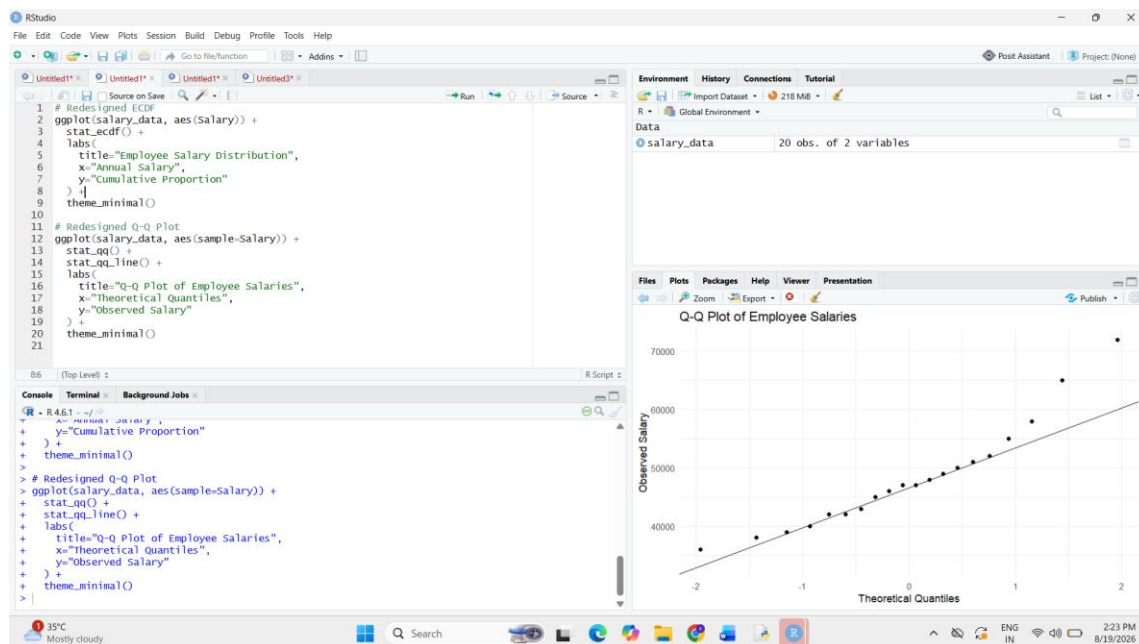
Assessment Task 3: Redesign

Redesigned ECDF

```
ggplot(salary_data, aes(Salary)) +  
  stat_ecdf() +  
  labs(  
    title="Employee Salary Distribution",  
    x="Annual Salary",  
    y="Cumulative Proportion"  
  ) +  
  theme_minimal()
```

Redesigned Q-Q Plot

```
ggplot(salary_data, aes(sample=Salary)) +  
  stat_qq() +  
  stat_qq_line() +  
  labs(  
    title="Q-Q Plot of Employee Salaries",  
    x="Theoretical Quantiles",  
    y="Observed Salary"  
  ) +  
  theme_minimal()
```



Assessment Task 4: Interpretation

The redesigned ECDF clearly shows cumulative proportions. The Q-Q plot with a reference line makes deviations from normality easier to identify. The **72,000 salary** appears as a high-end value and contributes to the upper-tail deviation.

Assessment Task 5: Is Q-Q plot alone sufficient?

No. A Q-Q plot alone is not sufficient. It should be supported by other methods such as a histogram, ECDF, summary statistics or a normality test.

2) Question 2 – Critique and Redesign Many Distributions Using Bean Plots

The dataset contains **20 students** in Python, Statistics, DataViz and AI.

Assessment Task 1: Original visualization

```
library(ggplot2)
```

```
student_data <- data.frame(
```

```
  Course=rep(c("Python", "Statistics", "DataViz", "AI"), each=5),
```

```
  Score=c(
```

62,68,74,81,88,

58,64,71,79,86,

55,61,69,77,84,

65,72,78,87,93

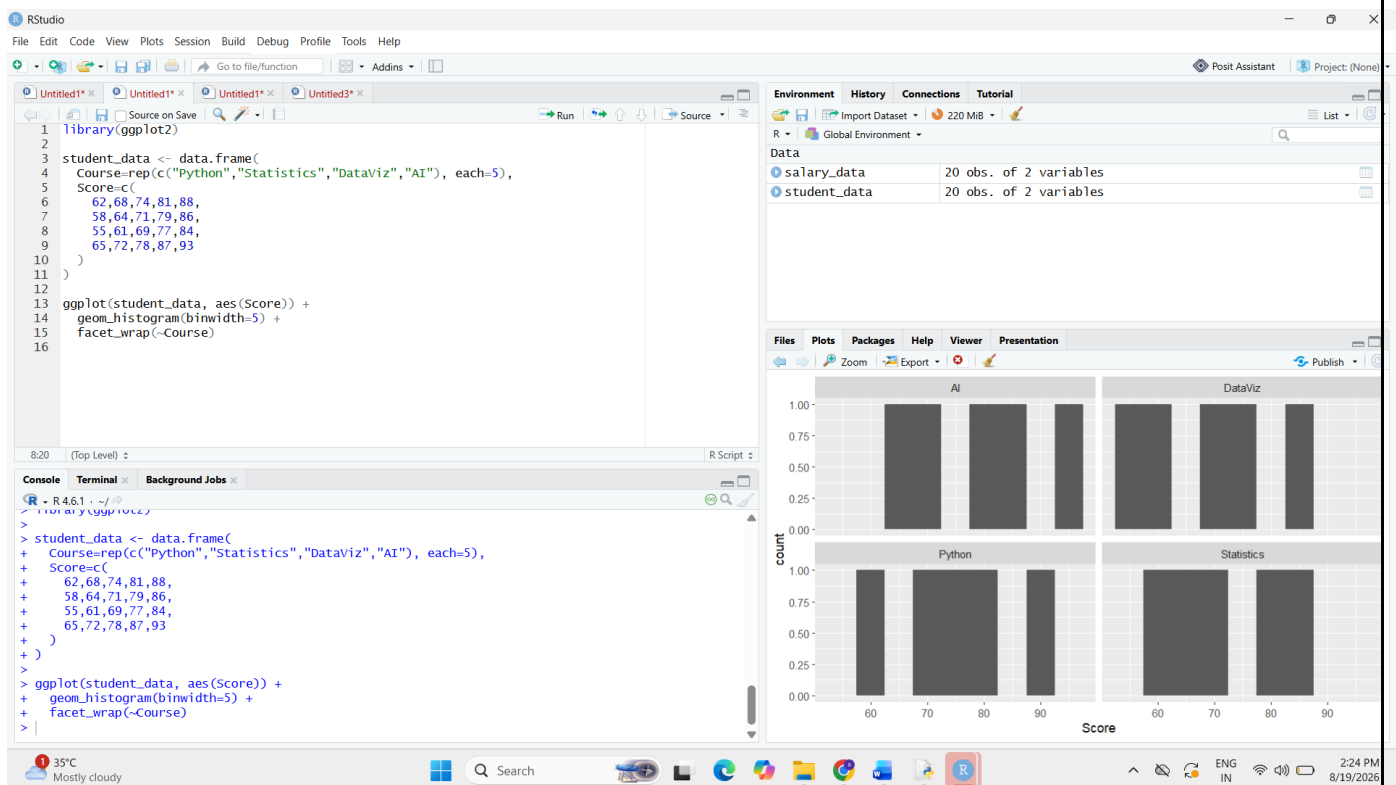
)

)

```
ggplot(student_data, aes(Score)) +
```

```
geom_histogram(binwidth=5) +
```

```
facet_wrap(~Course)
```



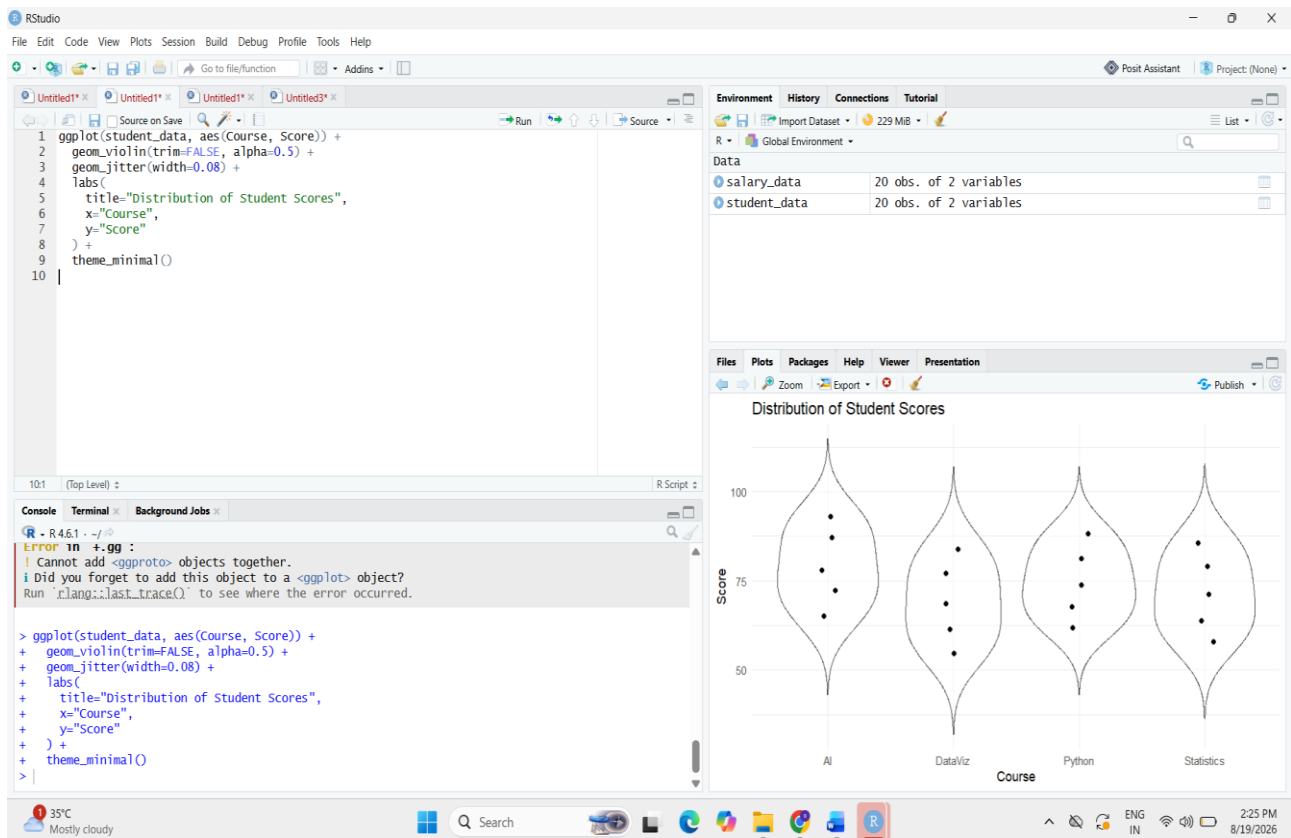
Assessment Task 2: Problems

1. Separate histograms make comparison difficult.
2. Different bin widths can distort distribution shape.

3. Scales and layouts are inconsistent.
4. Small samples make histograms less reliable.

Assessment Task 3: Redesign

```
ggplot(student_data, aes(Course, Score)) +  
  
  geom_violin(trim=FALSE, alpha=0.5) +  
  
  geom_jitter(width=0.08) +  
  
  labs(  
  
    title="Distribution of Student Scores",  
  
    x="Course",  
  
    y="Score"  
  ) +  
  
  theme_minimal()
```



Assessment Task 4: Common scale and labels

Use a common **50–100 score scale** with clear labels: **Python, Statistics, DataViz and AI**.

Assessment Task 5: Interpretation

- **Highest concentration:** Python
- **Greatest spread:** DataViz

Python scores range from **62–88**, while DataViz ranges from **55–84**. Therefore, DataViz has the greatest spread.

3) Question 3 – Critique and Redesign Proportion Visualizations

The dataset contains Laptop, Phone and Tablet purchases across **4 regions**.

Assessment Task 1: Original pie charts

```
purchase_data <- data.frame(  
  Region=rep(c("North","South","East","West"), each=3),  
  Product=rep(c("Laptop","Phone","Tablet"),4),  
  Purchases=c(  
    25,35,20,  
    30,40,25,  
    28,32,18,  
    22,38,27  
  )  
)  
  
par(mfrow=c(2,2))
```



```
for(r in unique(purchase_data$Region)){

  d <- subset(purchase_data, Region==r)

  pie(d$Purchases,

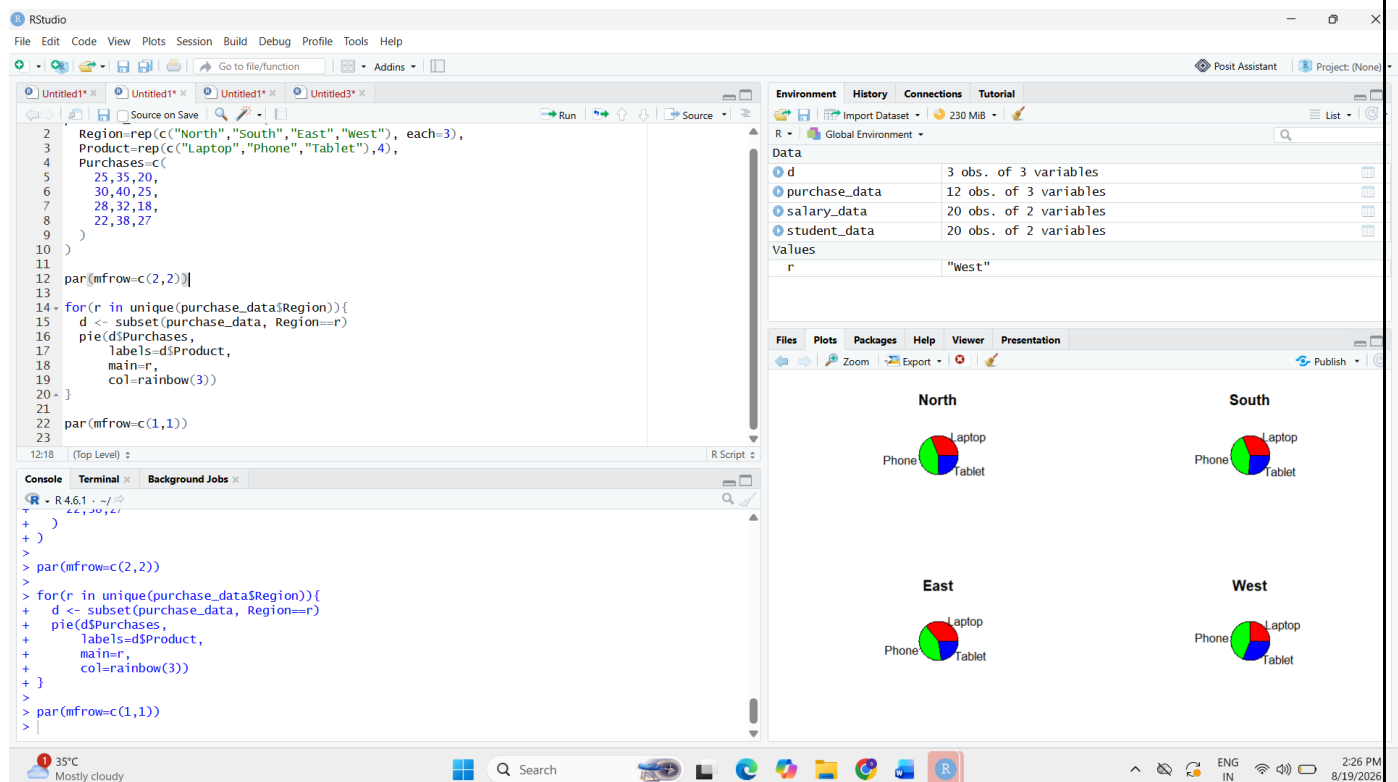
      labels=d$Product,

      main=r,

      col=rainbow(3))

}
```

```
par(mfrow=c(1,1))
```



Assessment Task 2: Problems

1. Pie charts are difficult to compare.
2. Different starting angles cause inconsistency.
3. 3-D effects can distort perception.

4. Inconsistent labels reduce clarity.

Assessment Task 3: Redesign

```
library(ggplot2)
```

```
# Side-by-side bars
```

```
ggplot(purchase_data,  
       aes(Region, Purchases, fill=Product)) +  
  geom_col(position="dodge") +  
  labs(  
    title="Product Purchases by Region",  
    x="Region",  
    y="Purchases"  
  ) +  
  theme_minimal()
```

```
# 100% stacked bars
```

```
library(dplyr)
```

```
p <- purchase_data %>%  
  group_by(Region) %>%  
  mutate(Proportion=Purchases/sum(Purchases))  
  
ggplot(p, aes(Region, Proportion, fill=Product)) +  
  geom_col() +
```

```
labs(

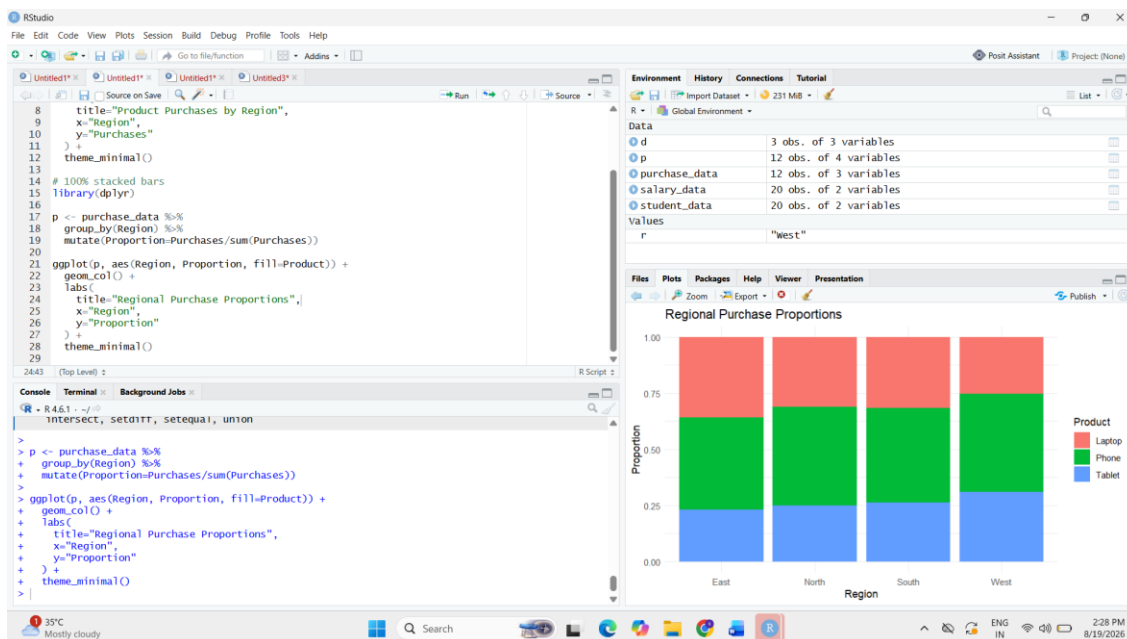
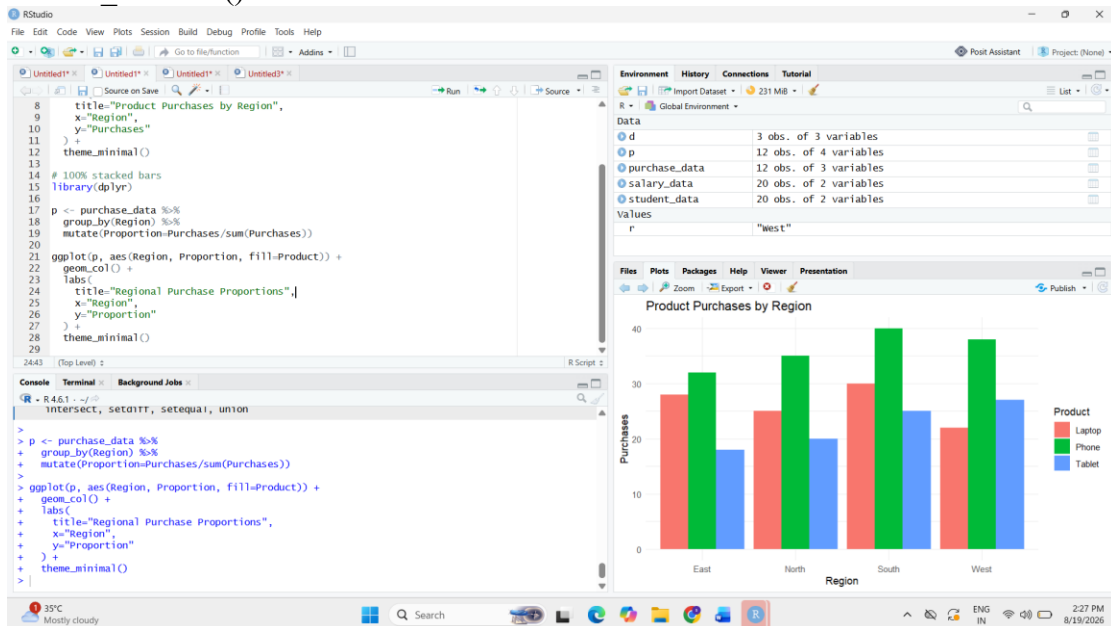
  title="Regional Purchase Proportions",

  x="Region",

  y="Proportion"

) +

theme_minimal()
```



Assessment Task 4: Comparison

Side-by-side bars: better for comparing actual purchase counts.

100% stacked bars: better for comparing regional proportions.

Therefore, **100% stacked bars are better for this task.**

Assessment Task 5: Why are 3-D effects misleading?

3-D effects and inconsistent scales distort the apparent size of slices, making values difficult to compare accurately.

4) Question 4 – Critique and Redesign Nested Proportions

The dataset represents **Region** → **Category** → **Subcategory** → **Sales**.

Assessment Task 1: Original treemap

```
library(treemap)
```

```
sales_data <- data.frame(  
  Region=rep(c("North","South","East","West"),each=4),  
  Category=rep(c("Electronics","Electronics",  
    "Furniture","Furniture"),4),  
  Subcategory=rep(c("Laptop","Phone","Chair","Desk"),4),  
  Sales=c(  
    52000,38000,22000,18000,  
    46000,42000,25000,21000,  
    58000,35000,19000,23000,  
    49000,44000,27000,20000
```

```

)

)

treemap(

  sales_data,

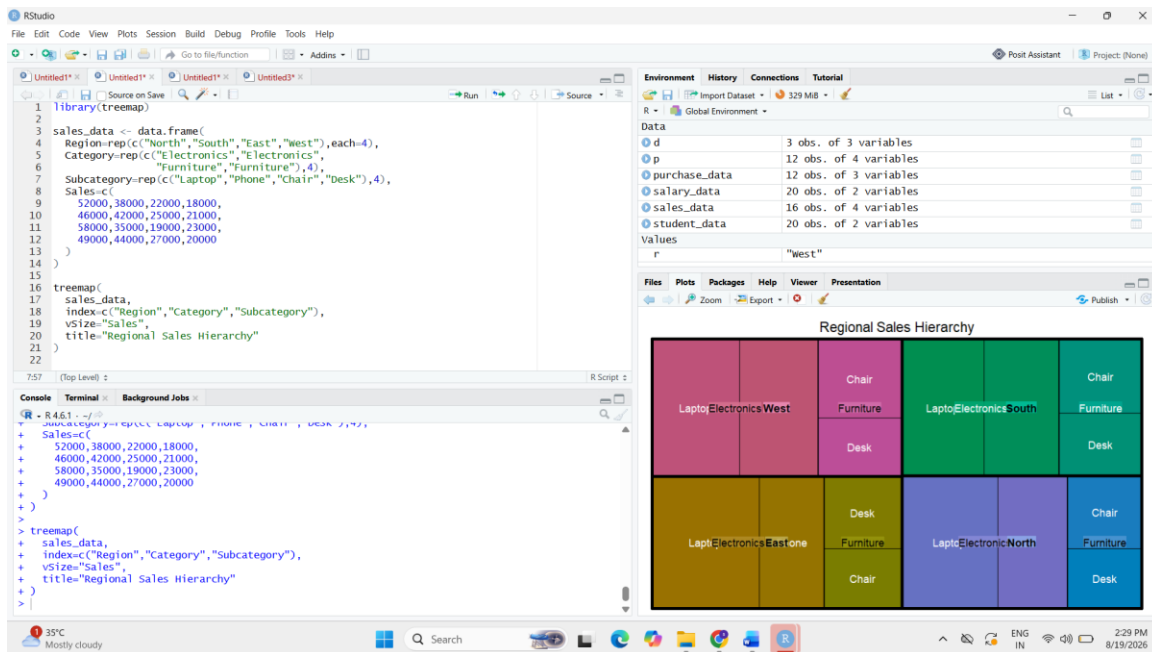
  index=c("Region","Category","Subcategory"),

  vSize="Sales",

  title="Regional Sales Hierarchy"

)

```



Assessment Task 2: Problems

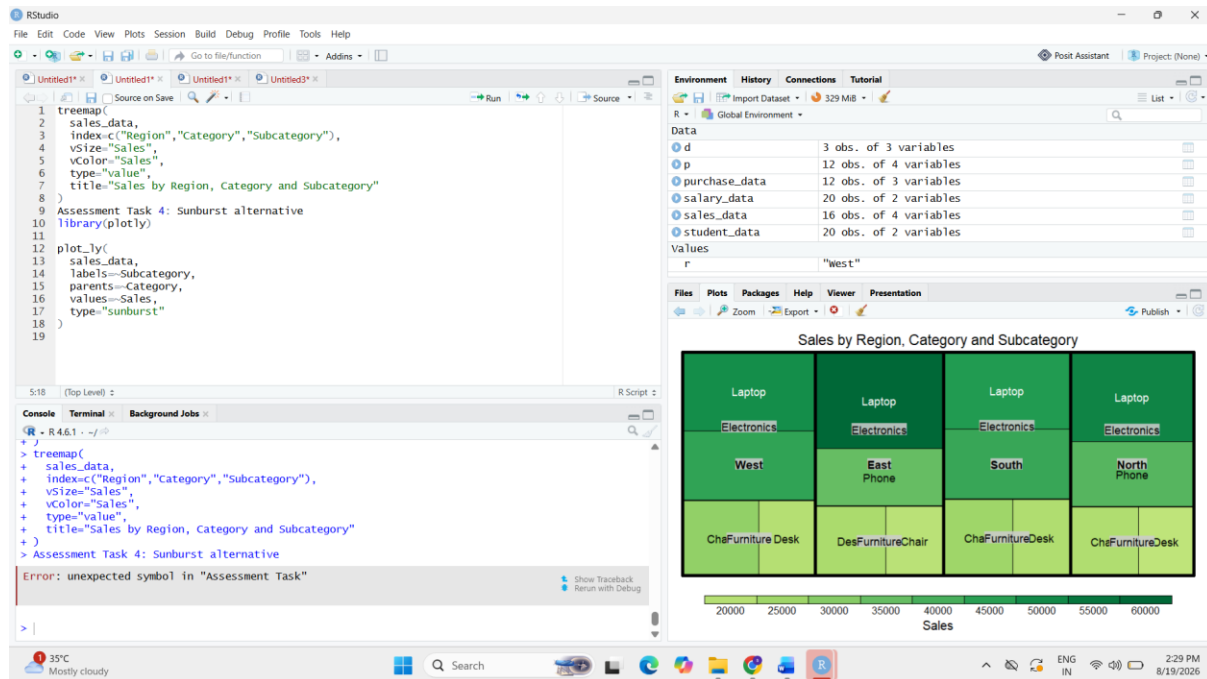
1. Small areas have unreadable labels.
2. Too many labels create clutter.
3. Colour does not clearly communicate hierarchy.
4. Similar colours make categories difficult to distinguish.

Assessment Task 3: Improved treemap

```
treemap(  
  sales_data,  
  index=c("Region","Category","Subcategory"),  
  vSize="Sales",  
  vColor="Sales",  
  type="value",  
  title="Sales by Region, Category and Subcategory"  
)
```

Assessment Task 4: Sunburst alternative

```
library(plotly)  
  
plot_ly(  
  sales_data,  
  labels=~Subcategory,  
  parents=~Category,  
  values=~Sales,  
  type="sunburst"  
)
```



Feature	Treemap	Sunburst
Sales comparison	Better	Good
Hierarchy	Good	Better
Space efficiency	Better	Good
Management use	Better	Good

Assessment Task 5: Recommendation

Improved treemap is recommended because management can quickly compare sales using area while viewing the Region → Category → Subcategory hierarchy.

5) Question 5 – Critique and Redesign Flow-Based Proportions

The dataset represents **Specialization → Internship → Placement Sector → Students**.

Assessment Task 1: Sankey diagram

```
library(ggplot2)
```

```
library(ggalluvial)
```

```
career_data <- data.frame(
```

```
  Specialization=c(
```

```
    "Data Analytics","Data Analytics","Data Analytics",
```

```
    "AI","AI","AI",
```

```
    "Cyber Security","Cyber Security","Cyber Security",
```

```
    "Cloud Computing","Cloud Computing","Cloud Computing"
```

```
  ),
```

```
  Internship=c(
```

```
    "Yes","Yes","No",
```

```
    "Yes","Yes","No",
```

```
    "Yes","Yes","No",
```

```
    "Yes","Yes","No"
```

```
  ),
```

```
  Placement_Sector=c(
```

```
    "IT","Finance","IT",
```

```
    "IT","Healthcare","IT",
```

```
    "IT","Finance","Government",
```

```
    "IT","Finance","Government"
```

```
  ),
```

```
  Students=c(
```



```
18,12,8,  
20,10,6,  
16,9,7,  
15,8,5  
)  
)  
  
ggplot(  
  career_data,  
  aes(  
    axis1=Specialization,  
    axis2=Internship,  
    axis3=Placement_Sector,  
    y=Students  
  )  
) +  
  geom_alluvium(aes(fill=Specialization)) +  
  geom_stratum() +  
  geom_text(  
    stat="stratum",  
    aes(label=after_stat(stratum))  
  ) +  
  labs(  
    title="Student Career Flow",
```

```

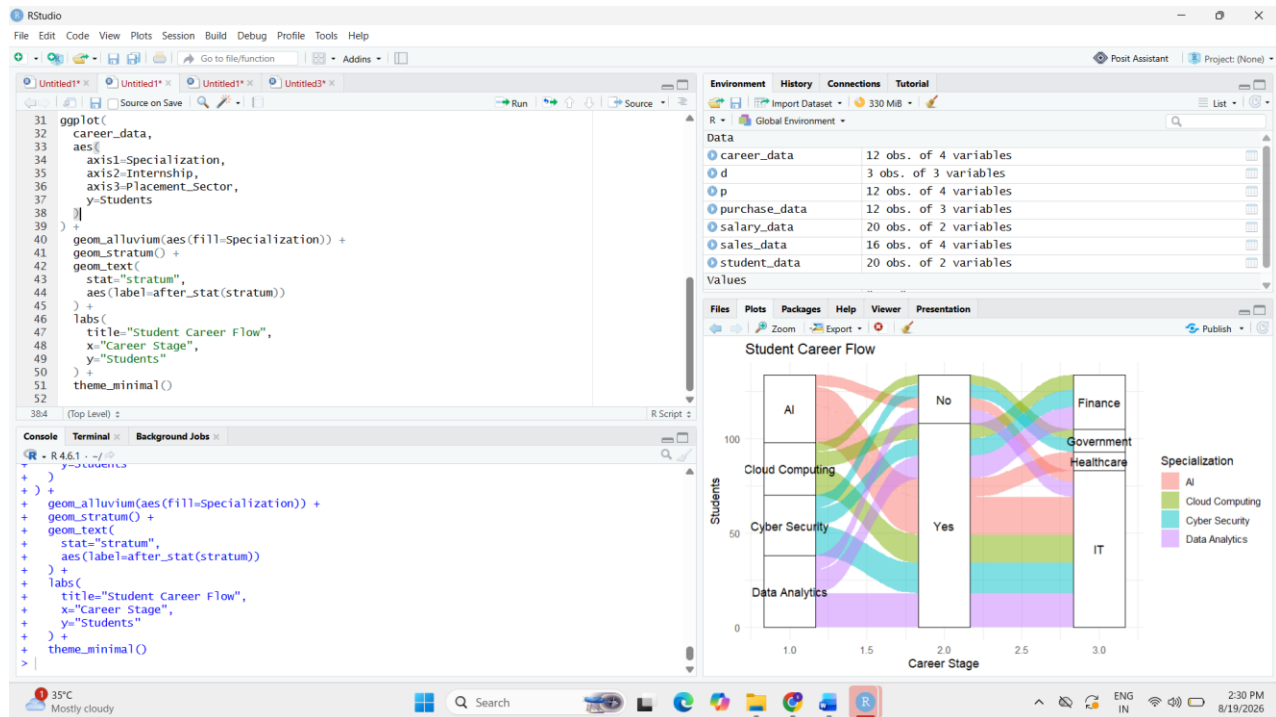
x="Career Stage",

y="Students"

)+

theme_minimal()

```



Assessment Task 2: Problems

1. Similar colours make flows difficult to distinguish.
2. Unlabelled flows are confusing.
3. Flow proportions are difficult to compare.
4. Small flows may be overlooked.
5. Crossing flows can create visual clutter.

Assessment Task 3: Redesign

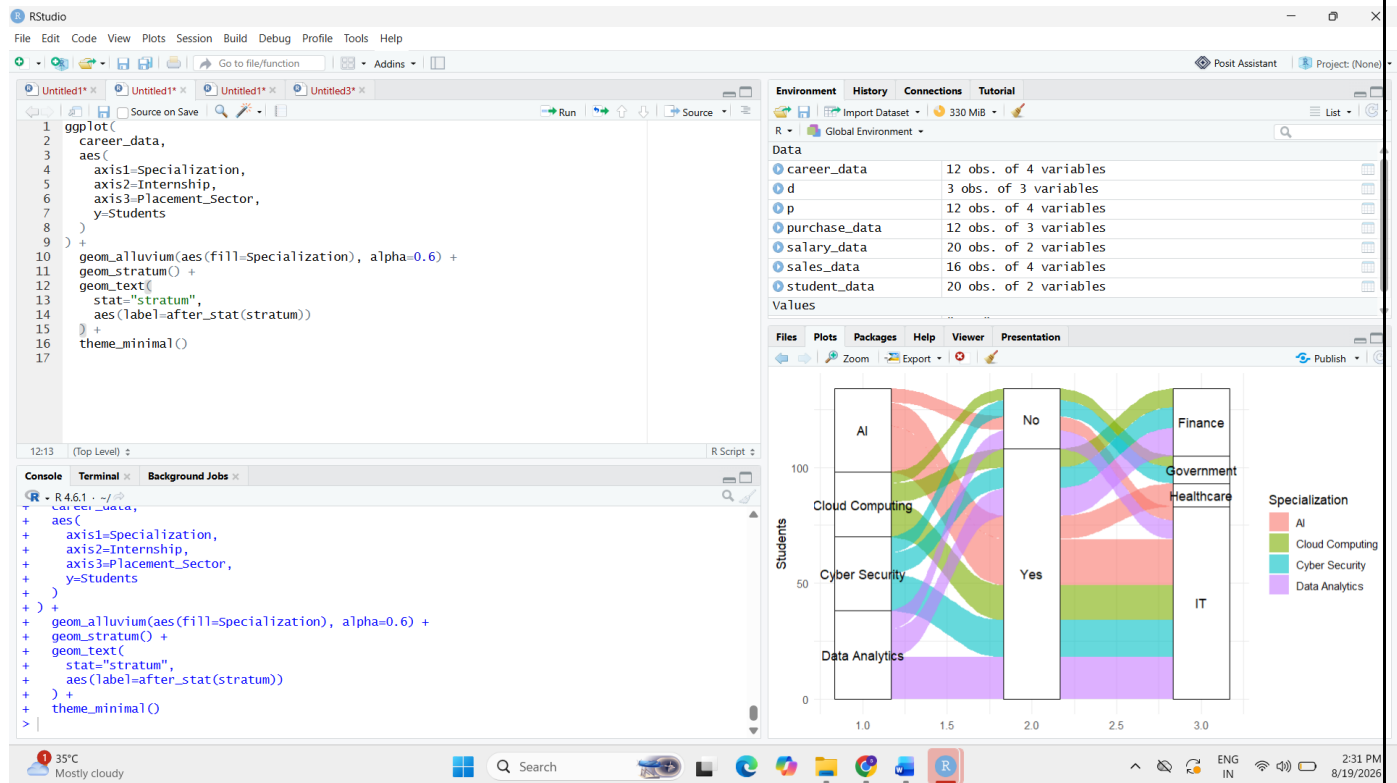
The redesigned Sankey uses:

- Clear node labels.

- Consistent flow widths.
- Specialization-based grouping.
- Clear stage labels.
- Minimal visual clutter.

Assessment Task 4: Parallel sets alternative

```
ggplot(  
  career_data,  
  aes(  
    axis1=Specialization,  
    axis2=Internship,  
    axis3=Placement_Sector,  
    y=Students  
  )  
) +  
  geom_alluvium(aes(fill=Specialization), alpha=0.6) +  
  geom_stratum() +  
  geom_text(  
    stat="stratum",  
    aes(label=after_stat(stratum))  
  ) +  
  theme_minimal()
```



Assessment Task 5: Recommendation

Sankey diagram is better because it clearly communicates movement from specialization to internship and finally to placement sector.

The largest flows are:

- **AI → Yes → IT = 20 students**
- **Data Analytics → Yes → IT = 18 students**
- **Cyber Security → Yes → IT = 16 students**
- **Cloud Computing → Yes → IT = 15 students**

Therefore, the **redesigned Sankey diagram** best communicates the major student flows.